ENG3014 _ Lab3: Eye Diagrams, Noise and Bandwidth Effects

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Introduction

This experiment investigates the behaviour of digital communication waveforms in the presence of noise, timing uncertainty and bandwidth limitations. Time domain waveforms were generated using PicoScope 6 and observed using a Keysight DSOX1102G oscilloscope. Eye diagrams were obtained using infinite persistence, allowing visualisation of intersymbol interference (ISI), jitter and noise. Both NRZ and PAM-4 formats were analysed.

1 NRZ Eye Diagram (Clean Signal):

A clean NRZ waveform was generated at $\approx 86\ kbit/s$. The eye diagram shows a single well defined opening with sharp transitions.

Vertical cursors were placed at the eye centre:

$$X_1 = 36.76\ \mu s, \qquad X_2 = 48.38\ \mu s$$

$$\Delta X = 11.62\ \mu s \qquad \Rightarrow \qquad R_b \approx \frac{1}{\Delta X} = 86.06\ kbit/s$$

Horizontal cursors gave:

$$Y_1 = -0.100\ V, \qquad Y_2 = +1.900\ V$$

$$\Delta Y = 2.00\ V$$

Thus, the clean NRZ eye is wide and tall, indicating excellent timing margin and noise immunity. Figure 1.

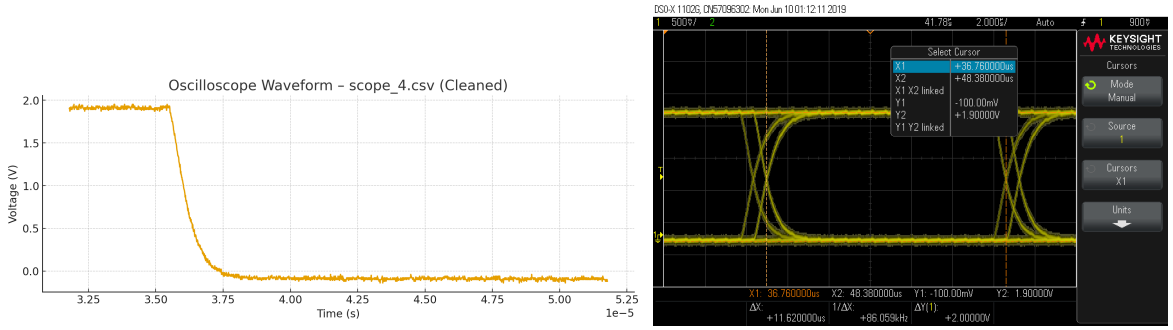


Figure 1: Time-domain waveform captured from the Keysight DSOX1102G

2 Trigger and Jitter Analysis:

Edge triggering was used while varying the trigger threshold. When the trigger level was set too close to regions of high noise, the oscilloscope triggered inconsistently, causing increased horizontal spread (timing jitter) in the eye diagram. Therefore, changing the trigger value changes how sensitive the trigger point is to noise, leading to different amounts of jitter.

Best performance occurred when the trigger was set near the midpoint of the rising edge, where:

$$Jitter = X_2 - X_1 = 0.68\ ns$$

At incorrect trigger levels the jitter increased significantly. This demonstrates that improper triggering can distort time domain measurements even if the signal itself is unchanged. Look Figure 2. When the input

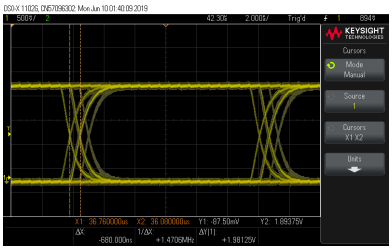


Figure 2: Eye diagram of the NRZ cleare signal 24dB (No Noize)

frequency was increased from 50 Hz to 300 Hz, the shape of the eye diagram did not significantly change. The width and height of the eye remained almost constant. This is because the eye diagram represents the timing of digital symbol transitions, not the frequency of an analogue sine wave. Therefore the symbol rate cannot be determined from these changes, and the observed 'eye' is not linked to the frequency selected in PicoScope. The reason is that a sine wave does not contain discrete digital symbols, so the oscilloscope overlays identical analogue cycles, producing an eye-like pattern that does not reflect the actual symbol rate of a data stream. As in Figure 3.

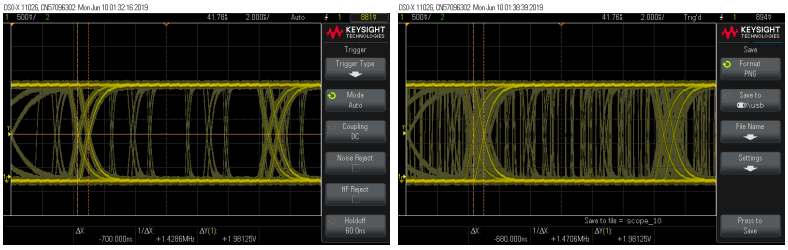


Figure 3: Testing frequency's from 50 Hz to 300 Hz

3 Height and Width of The Eye:

SNR and Noise Calculation Overview

Before generating signals with different (SNRs), we first need to measure the RMS voltage of the clean signal. This allows us to compute the exact noise level required for each SNR value. Since SNR in decibels is defined using a power ratio, and power is proportional to the square of the RMS voltage, we can convert $\mathrm{SNR_{dB}}$ to a voltage ratio. Using the measured signal RMS value, we then calculate the corresponding noise RMS voltages for the selected SNR values. These results are summarised in the Table 1 below. $\mathrm{V}_{noise,rms} = \frac{V_{signal,rms}}{10^{\mathrm{SNR_{dB}}/20}}$.

SNR (dB)	$\frac{V_{signal}}{V_{noise}}$	$V_{noise,rms}$ (V)
24	$10^{24/20} = 15.85$	0.0548
18	$10^{18/20} = 7.94$	0.1093
12	$10^{12/20} = 3.98$	0.2182
8	$10^{8/20} = 2.51$	0.3452
6	$10^{6/20} = 1.995$	0.4352

Table 1: Calculated noise RMS voltages using measured $V_{signal,rms} = 0.8684$ V.

Random noise was added at several SNR values (18, 12, 8, 6 dB). As expected, decreasing SNR caused:

- reduced vertical eye height,
- thicker noise bands on logical "0" and "1" levels,
- increased probability of error at the sampling instant.

At 6 dB SNR, although the NRZ eye narrowed, a detectable eye opening was still present. This shows NRZ has relatively strong noise robustness.

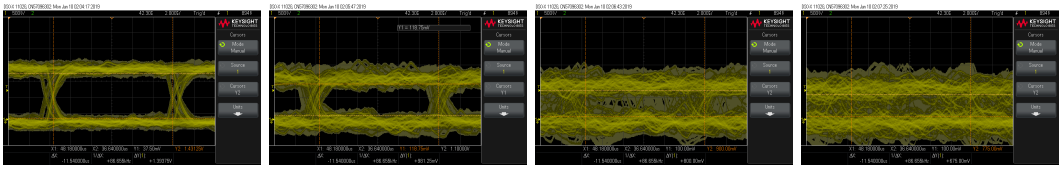


Figure 4: Eye diagram with AWGN at SNR values (18, 12, 8, 6 dB)

4 Bandwidth Limitation (Low-Pass Filter)

A digital low-pass filter was applied using the oscilloscope's `Math → Low Pass Filter` function.

- At 10 MHz bandwidth, the filtered waveform closely matched the original, with an open eye.
- At 40 kHz, bandwidth Limitation occurs, when the bandwidth was reduced to 40 kHz, the filtered waveform became extremely smoothed. because high-frequency components required to create sharp transitions were removed,filtered trace became so slow and attenuated that the Math display temporarily disappeared from the oscilloscope screen.

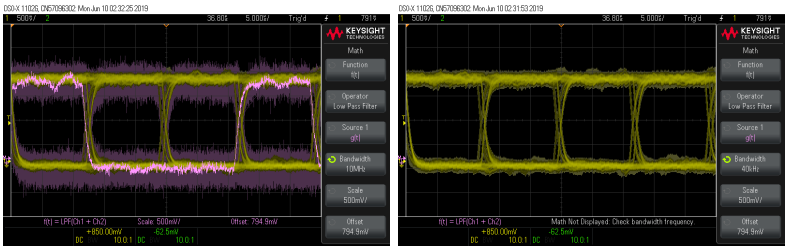


Figure 5: Lift: LPF 10 MHz , Right: 40 kHz wave disappeared

5 PAM-4 Eye Diagram (Clean Signal)

A clean PAM-4 signal was then generated at ≈ 55 $ksymbols/s$. Three eye openings are visible, corresponding to the transitions:

$$Level\ 0 \leftrightarrow 1, \qquad 1 \leftrightarrow 2, \qquad 2 \leftrightarrow 3.$$

Measured using oscillocope cursors:

$$H_{top=mid=bot} = 0.6625\ V,$$

Thus, all three eyes have nearly equal height and spacing, as expected for an ideal PAM-4 modulator. The symbol period was:

$$\Delta X = 18.20\ \mu s \qquad \Rightarrow \qquad R_s = 54.95\ ksymbols/s$$

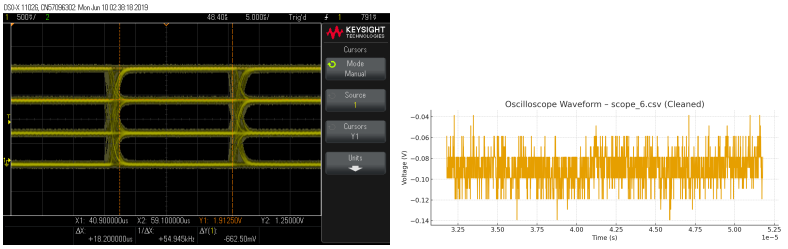


Figure 6: Lift: PAM-4 24dB Eye Diagram , Right: 6dB Time Domain

6 Effect of Noise on PAM-4

Three noisy PAM-4 signals were examined (18 dB, 12 dB, 6 dB SNR):

18 dB SNR: The eye openings reduced to approximately:

$$H_{18dB} \approx 0.425\ V$$

Rails thickened noticeably, but decision boundaries remained visible.

12 dB SNR: Noise became comparable to the level spacing. Eyes partially collapsed and symbol levels overlapped, especially in the middle region. Reliable detection was difficult.

6 dB SNR: Near complete eye closure occurred. All four levels merged, no distinct decision regions remained. Symbol recovery becomes impossible at this SNR.

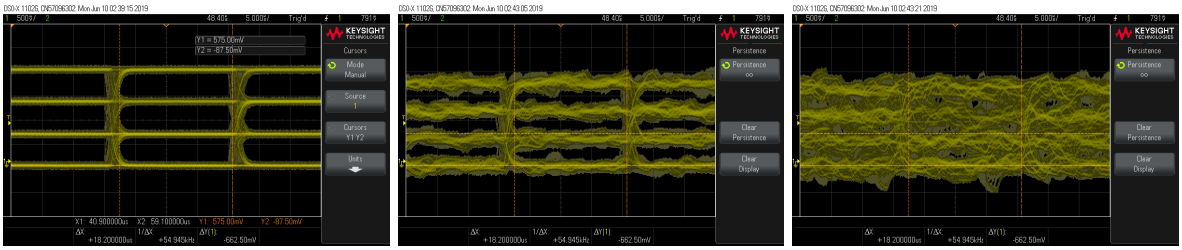


Figure 7: PAM-4 from L to R :18,12,6 dB Eye Diagram

6.1 Final Connection to the Theory

Shannon Hartley stases:
The number of signal levels can reliably encoded is fundamentally limited by SNR.
PAM-4 requires enough SNR to keep four decision regions separate.
*From the Experiment: As SNR drops, the channel cannot support multiple voltage levels.

7 Discussion

PAM-4 is significantly more sensitive to noise than NRZ. Because PAM-4 encodes two bits per symbol using four closely spaced voltage levels, the eye openings are inherently smaller. Noise therefore causes much greater vertical and horizontal closure. NRZ remains detectable at low SNR, whereas PAM-4 collapses at 6 dB.

Bandwidth limitation also dramatically affects both modulations, demonstrating the trade off between symbol rate, ISI and channel frequency response.

8 Conclusion

This experiment demonstrated the practical behaviour of communication waveforms under noise, timing uncertainty and filtering constraints. Eye diagrams provide a powerful visual tool for evaluating system performance, showing that:

- NRZ has strong noise tolerance.
- PAM-4 requires high SNR to maintain reliable detection.
- poor trigger choice increases jitter.
- limited channel bandwidth causes ISI and eye closure.

These observations match theoretical expectations from digital communication theory.